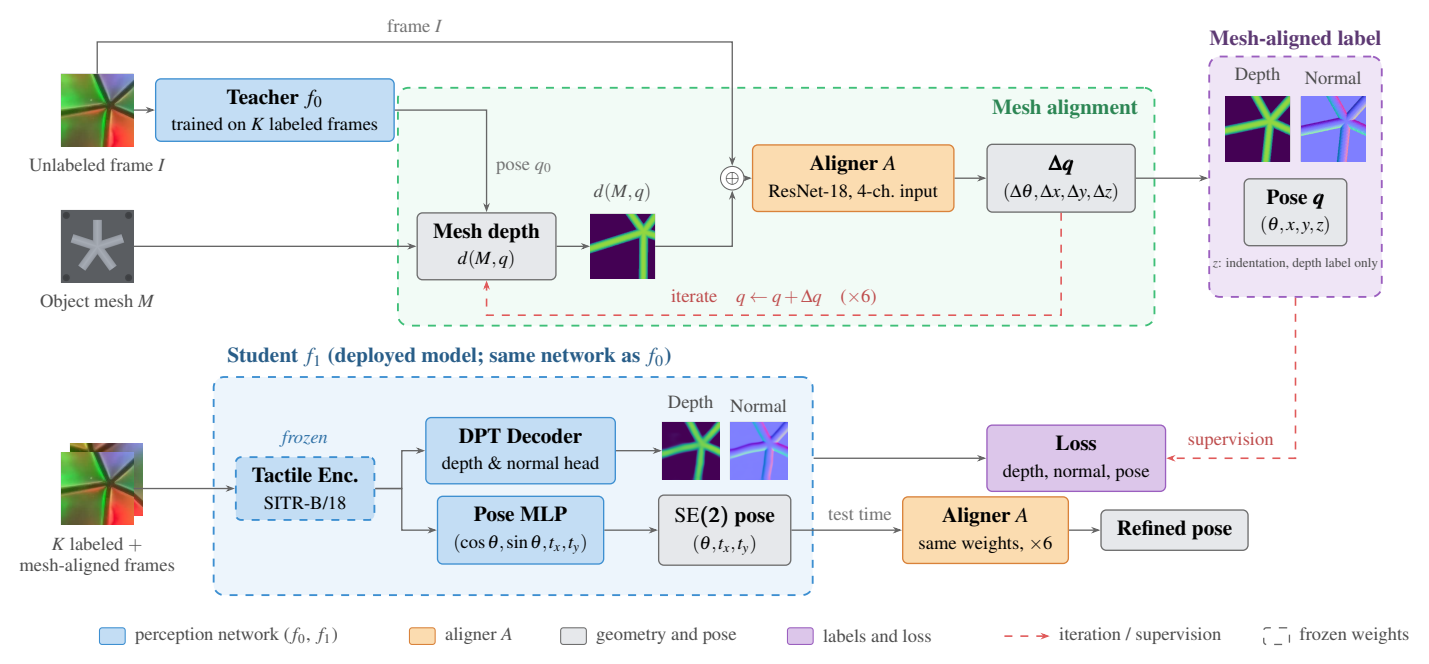

frame $I$
Unlabeled frame $I$
Teacher $f_0$
trained on $K$ labeled frames
Object mesh $M$
pose $q_0$
$d(M,q)$
Mesh depth
$d(M,q)$
Mesh alignment
Aligner $A$
ResNet-18, 4-ch. input
$\Delta q$
$(\Delta\theta, \Delta x, \Delta y, \Delta z)$
iterate $q \leftarrow q + \Delta q$ $(\times 6)$
Mesh-aligned label
Depth
Normal
Pose $q$
$(\theta, x, y, z)$
$z$: indentation, depth label only
Student $f_1$ (deployed model; same network as $f_0$)
$K$ labeled + mesh-aligned frames
frozen
Tactile Enc.
SITR-B/18
DPT Decoder
depth & normal head
Pose MLP
$(\cos\theta, \sin\theta, t_x, t_y)$
Depth
Normal
SE(2) pose
$(\theta, t_x, t_y)$
Loss
depth, normal, pose
supervision
test time
Aligner $A$
same weights, $\times 6$
Refined pose
perception network ($f_0$, $f_1$)
aligner $A$
geometry and pose
labels and loss
iteration / supervision
frozen weights